CoOp

Learning to Prompt for Vision-Language Models

IJCV 2022

CORE IDEA

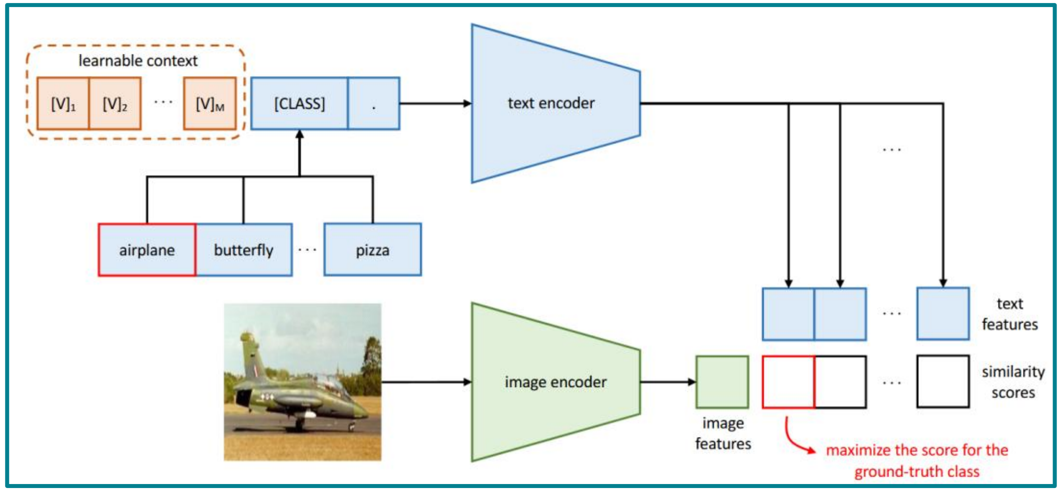
Replaces CLIP's hand-crafted text prompt ("a photo of a [CLASS]") with a set of learnable continuous context vectors, optimized on few-shot labels while both CLIP encoders stay frozen. Only the prompt is trained.

METHOD ANATOMY

$$t = [V]_1[V]_2 \ \ldots \ [V]_M \ [CLASS]$$

M learnable context tokens $V_i \in \mathbb{R}^d$ (d = token dim). Shared (unified) or class-specific. Optimized via cross-entropy on cos(image, text) logits.

ARCHITECTURE



SPEC SHEET

Tunes	Text prompt context vectors
Frozen	Both CLIP encoders (image + text)
Training-free?	No — gradient descent
# Params	M × d (e.g. 16×512 ≈ 8k)
Inference cost	Same as zero-shot CLIP

STRENGTHS / WEAKNESSES

- ➕ Large gains over zero-shot in low-shot regimes
- ➕ Tiny parameter footprint; encoder-agnostic
- ➖ Weak base→new generalization (overfits base)
- ➖ Sensitive to support-set quality & seed

Paper: Zhou et al. 2022 **Repo:** github.com/KaiyangZhou/CoOp